HSAN ELLOUZE

AI Engineering Student | Document AI, LLM Agents & Python Backends
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Seeking a 6-month end-of-studies internship (PFE) starting February 2027 | Open to relocation (Berlin)

Summary

Final-year Software Engineering student (INSAT, expected graduation September 2027) with production experience turning unstructured business documents into structured, usable data: PDF parsing and chunking pipelines, LLM-backed contract analysis for enterprise clients, and RAG systems. Has built LLM agents with LangChain/LangGraph and compared providers (OpenAI, Gemini, Ollama) on controlled scenarios. Grounded in the Python and backend engineering (FastAPI, REST APIs, Docker, CI/CD) needed to take an AI product into production for real customers.

Experience

AI Engineer Intern – Applicare, Canada Jun 2026 – Aug 2026
- Developed a dual-source **document ingestion** pipeline (PubMed XML + GROBID **PDF parsing**) with structure-aware (IMRAD) chunking and a sliding-window fallback for documents that do not follow the expected layout
- Architected AskSahera, a bilingual (FR/EN) production RAG chatbot indexing ~100 source documents
- Built the RAG backend in Express.js with Qdrant vector storage and OpenAI embeddings, using Claude for generation; integrated Cloudflare R2 for storage

Mazdev – Tunis
Software Engineer Intern Jul 2025 – Sep 2025
- Engineered a production chatbot platform (YAML-configurable) that let non-technical business users create 20+ bot variants without code
- Created a RAG engine (Pinecone) reaching 94% semantic accuracy across a large document set; optimized embeddings to cut latency 45%
- Established an automated testing framework, cutting the QA cycle 50%

Fullstack Developer Mar 2025 – Jun 2025
- Architected an **AI-powered document analysis** platform for legal contract review, used by 40+ enterprise clients and cutting analysis time 60%
- Delivered a React/Next.js + **FastAPI (Python)** REST API system processing 500+ contract reviews per month at production scale
- Implemented a Docker CI/CD pipeline, cutting infra setup time 75%

Frontend Developer Intern – Steros, Sfax Jul 2024 – Aug 2024
- Built a responsive showcase website and interactive UI components, with a PHP/MySQL backend

Projects

KubeMind: AI-Driven Kubernetes Observability Platform (3-person capstone team) GitHub
- Implemented 3 specialized **LLM agents** in LangChain/LangGraph (logs, metrics, events) orchestrated via a LangGraph StateGraph workflow, feeding a correlation agent
- Designed a 5-layer agentic software architecture (ingestion, AI agents, correlation/storage, presentation) for automated incident diagnosis
- Developed a FastAPI backend and Next.js dashboard on a real-time ingestion pipeline (Prometheus, Loki, Kafka, MongoDB)
- Benchmarked output quality across 3 LLM providers (OpenAI, Gemini, Ollama) across 3 controlled incident scenarios

Multi-Tenant AI Chatbot Widget GitHub
- Designed a reusable, embeddable AI chatbot component for multi-tenant deployment across client websites via a single `<script>` tag
- Enforced Shadow DOM isolation and strict domain validation for secure **multi-tenant** use
- Integrated the backend with Hono, Supabase, and Upstash Redis; deployed on Railway and Cloudflare Pages

Education

INSAT – National Institute of Applied Sciences and Technology Sep 2022 – Sep 2027 (expected)
Engineering Degree in Software Engineering (EQF Level 7 / Master's equivalent)

Skills

AI/ML: LLM Integration, RAG, LangChain, LangGraph, Embeddings, Prompt Engineering, Vector Databases (Qdrant, Pinecone)
Languages: Python, JavaScript, TypeScript, Java, C#, PHP, SQL
Backend: FastAPI (Python), Express.js (JavaScript), NestJS (JavaScript), Hono (TypeScript), ASP.NET Core (C#)
DevOps & Cloud: Docker, Kubernetes, GitHub Actions, CI/CD, Vercel, Railway, Cloudflare (Pages, R2)
Data & Observability: Redis (Upstash), MongoDB, MySQL, Supabase, Prometheus, Loki, Kafka
Frontend: React, Next.js, Vue
Tools: Git, Linux

Activities

Event Mission Leader – Junior Enterprise INSAT 2024 – 2025
- Supervised a ~40-member team over the 2024–2025 mandate, leading the organizing committee for 2 flagship events, including Forum INSAT Entreprise (1,500+ attendees, 40+ companies)

Operations Manager – WinterCup 7.0 2025
- Owned end-to-end logistics and event planning for WinterCup 7.0, an overnight coding contest with ~300 participants, coordinating the on-site operations staff under time pressure